Exercises2

2.1 Let $\mathcal{R}, \mathcal{S}, \mathcal{T}$ be subspaces of a finite dimensional linear space $\mathcal{X}$.

1. Show that in general

$$\mathcal{R} \cap (\mathcal{S} + \mathcal{T}) \neq \mathcal{R} \cap \mathcal{S} + \mathcal{R} \cap \mathcal{T},$$

   i.e., the lattice of all subspaces of $\mathcal{X}$ is not distributive.

2. Show that if $\mathcal{T} \subset \mathcal{R}$, then

$$\mathcal{R} \cap (\mathcal{S} + \mathcal{T}) = \mathcal{R} \cap \mathcal{S} + \mathcal{T}.$$

   i.e., the lattice of all subspaces of $\mathcal{X}$ is modular.

3. Show that if any one of the following is true, then all are true.

$$\begin{aligned} \mathcal{R} \cap (\mathcal{S} + \mathcal{T}) &= \mathcal{R} \cap \mathcal{S} + \mathcal{R} \cap \mathcal{T} \\ \mathcal{S} \cap (\mathcal{R} + \mathcal{T}) &= \mathcal{S} \cap \mathcal{R} + \mathcal{S} \cap \mathcal{T} \\ \mathcal{T} \cap (\mathcal{R} + \mathcal{S}) &= \mathcal{T} \cap \mathcal{R} + \mathcal{T} \cap \mathcal{S}. \end{aligned}$$

2.2 Verify (2.1-2.4) and also verify that (2.2) and (2.3) become equalities when (2.5) and (2.6) hold respectively. (The relevant equations for Exercise 2.2 are on page 17 of notes_v1).

2.5 Let $A : \mathcal{X} \to \mathcal{X}$ be a linear transformation and $\mathcal{U} \subset \mathcal{X}$ be a subspace. Prove that $A(A^{-1}\mathcal{U}) = \mathcal{U} \cap \mathcal{R}(A)$ and $A^{-1}(A\mathcal{U}) = \mathcal{U} + \mathcal{N}(A)$.

2.6 Assume A, B be matrices with the same number of rows. Prove that $\mathcal{R}(A) \subset \mathcal{R}(B)$ if and only if $\mathcal{N}(B^*) \subset \mathcal{N}(A^*)$. Also show the following equalities:

1. $\mathcal{R}(A) = \mathcal{N}(A^*)^{\perp}$.
2. $\mathcal{R}(A) = \mathcal{R}(AA^*)$ and $\mathcal{N}(A) = \mathcal{N}(A^*A)$.

2.13 Prove that the Hölder p-norms are indeed norms in $\mathbb{F}^n$. Prove that the Ky Fan k-norms are indeed norms in $\mathbb{F}^n$.